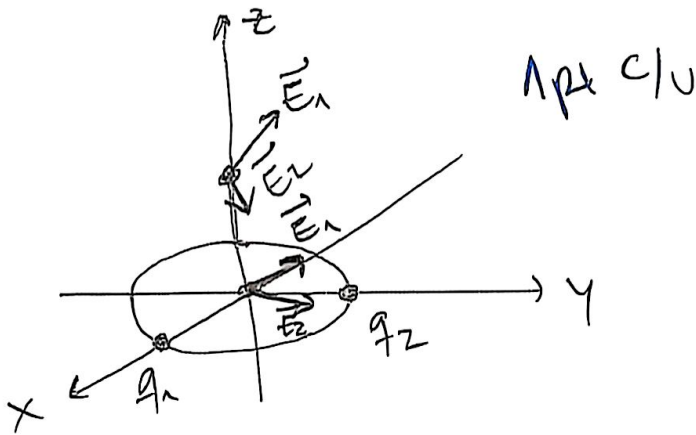


Parte V.1.

a)



b) $\vec{E}(P_1) = \vec{E}_1 + \vec{E}_2$ (1 pt)

$$\vec{E}_1 = \frac{k_e q_1 (\vec{r}_{P1} - \vec{r}_1)}{|\vec{r}_{P1} - \vec{r}_1|^3} = \frac{8,99 \cdot 10^9 \frac{Nm^2}{C^2} \cdot 12 \cdot 10^{-6} C \cdot (-0,2 \hat{x} m)}{8 \cdot 10^{-3} m^3}$$

$$\vec{r}_{P1} = \vec{0} m$$

$$\vec{r}_1 = 0,2 \hat{x} m$$

$$\vec{r}_{P1} - \vec{r}_1 = -0,2 \hat{x} m$$

$$|\vec{r}_{P1} - \vec{r}_1|^3 = 8 \cdot 10^{-3} m^3$$

$$\vec{E}_1 = -2697000 \hat{x} \frac{N}{C} \quad (1,5)$$

1.

$$\vec{E}_2 = \frac{k_e q_2 (\vec{r}_{P1} - \vec{r}_2)}{|\vec{r}_{P1} - \vec{r}_2|^3} = \frac{8,99 \cdot 10^9 \frac{Nm^2}{C^2} \cdot (-6 \cdot 10^{-6} C) \cdot (-0,2 \hat{y} m)}{8 \cdot 10^{-3} m^3}$$

$$\vec{r}_{P1} = \vec{0} m$$

$$\vec{r}_2 = 0,2 \hat{y} m$$

$$\vec{r}_{P1} - \vec{r}_2 = -0,2 \hat{y} m$$

$$|\vec{r}_{P1} - \vec{r}_2|^3 = 8 \cdot 10^{-3} m^3$$

$$\vec{E}_2 = 1348500 \hat{y} \frac{N}{C} \quad (1,5)$$

1.

$$\vec{E} = -2697000 \hat{x} + 1348500 \hat{y} \frac{N}{C} \quad (1 pt)$$

$$b) \vec{E}(P_2) = \vec{E}_1 + \vec{E}_2 \quad (1pt)$$

$$\vec{E}_1 = \frac{k_e q_1 (\vec{r}_{P2} - \vec{r}_1)}{|\vec{r}_{P2} - \vec{r}_1|^3} = \frac{8,99 \cdot 10^9 \frac{Nm^2}{C^2} \cdot 12 \cdot 10^{-6} C \cdot (-0,2\hat{x} + 0,3\hat{k})}{0,047 m^3}$$

$$\begin{cases} \vec{r}_{P2} = 0,3\hat{k} m \\ \vec{r}_1 = 0,2\hat{x} m \\ \vec{r}_{P2} - \vec{r}_1 = -0,2\hat{x} + 0,3\hat{k} m \\ |\vec{r}_{P2} - \vec{r}_1|^3 = \sqrt{(-0,2)^2 + (0,3)^2}^3 = 0,047 m^3 \end{cases} \quad \vec{E}_1 = -459063,8\hat{x} + 688595,7\hat{j} \frac{N}{C} \quad (1,5)$$

$$\vec{E}_2 = \frac{k_e q_2 (\vec{r}_{P2} - \vec{r}_2)}{|\vec{r}_{P2} - \vec{r}_2|^3} = \frac{8,99 \cdot 10^9 \frac{Nm^2}{C^2} \cdot (-6 \cdot 10^{-6} C) (-0,2\hat{y} + 0,3\hat{k})}{0,047 m^3}$$

$$\begin{cases} \vec{r}_{P2} = 0,3\hat{k} m \\ \vec{r}_2 = 0,2\hat{y} m \\ \vec{r}_{P2} - \vec{r}_2 = -0,2\hat{y} + 0,3\hat{k} m \\ |\vec{r}_{P2} - \vec{r}_2|^3 = \sqrt{(-0,2)^2 + (0,3)^2}^3 = 0,047 m^3 \end{cases} \quad \vec{E}_2 = 229531,9\hat{j} - 344297,9\hat{k} \frac{N}{C} \quad (1,5)$$

$$\vec{E}_T = -459063,8\hat{x} + 918127,6\hat{j} - 344297,9\hat{k} \frac{N}{C} \quad (1)$$